

Carbon Calculator Service.java.

Emission Calculated.

Added required dependencies in pom.xml

- Hibernate ORM
- MySQL Connector
- Jakarta Persistence.

Entity classes (DB Mapping)

Created Hibernate entity classes

User.java maps to user table.

Activity. " Activity "

Carbon-Record. " Carbon record table "

Used Annotations

@ Entity

@ Table

@ id

@ Generated Value

@ column

Current Project Status

- ✓ User tracking
- ✓ Activity logging
- ✓ Carbon emissions calculation
- ✓ Database integration

Hibernate Configuration

Created : hibernate.cfg.xml.

Configured : DB URL Username
 MySQL driver password
 Hibernate properties

Repository layer (DB operations)

User saveUser() Act - SaveActivity() for SaveRecord
 getAllUsers()

Service layer (Business Logic)

U registerUser()
 getUsers()

A addActivity()

calculate CO₂ emis.

formula → CO₂ = activity value × emission factor

Application Entry Point.

functions like.

Add User

Read User

Add Activity

emission